PRODUCT MANAGER TAKE-HOME · NOMAD VENTURE STUDIO / JOBESCAPE

Appendix

The working: generated analysis output, every query, and the verification run.

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Contents

What is in this appendix

Everything here is **generated output**, not hand-written prose. If any figure in the submission, the dossier or the deck is ever questioned, the answer is in one of these files — and all of it can be regenerated from the source CSVs in one command.

```
cd part-c-release-verdict/analysis
python3 05_qa.py          # re-derives all 91 published figures and fails loudly on any drift
```

Contents

FILE	WHAT'S IN IT	ANSWERS
`05_qa-output.txt`	**91 assertions**, each computed from the raw CSV and checked against the published figure	*"Is this number real?"* — start here
`02_main-output.txt`	The full Part C Task-1 analysis: exposure ladder, tier calibration, the headline comparison with confidence intervals, all four causal tests, threshold-robustness sweep, and every segment cut	*"Show your calculations"*
`03_supplement-output.txt`	Target-metric definitions side by side, the CSAT inversion, the refund-gate test, the popup-decliner group, data-quality flags, and the economic sizing of the feature	*"What else did you check?"*
`04_loose_ends-output.txt`	Cohort baselines, the confound check on exposure, the **stratified takers-vs-control comparison**, challenge-start dates, and the plan-mix validation against Part D	*"Is the effect just activity level?"*
`01_diagnostics-output.txt`	The questions settled *before* the engagement tiers were chosen: challenge-start lag, the 8-lesson structure, dose-response by both measures, censoring, and coverage of course 338	*"How did you pick the cut?"*
`sql/`	All 8 BigQuery queries — see the table below for which produced data	*"How did you pull it?"*

The queries, and which ones are load-bearing

`00` probes the real schema; `01` – `03` were written **before** it was known, against assumed column names, and were superseded rather than run — they are kept because they are the actual working, and they still carry their `⚠ADJUST` markers. Five queries produced every CSV the analysis reads.

QUERY	RAN	PRODUCED
`00_explore_schema.sql`	✅	95 rows — the real column list for both tables
`01_engagement_tiers.sql`	—	superseded draft (assumed schema)
`02_core_metrics.sql`	—	superseded draft (assumed schema)
`03_segment_breakdown.sql`	—	superseded draft (assumed schema)
`10_explore_events.sql`	✅	the 67-event catalogue, event×course counts, unsubscribe-reason counts, row samples

QUERY	RAN	PRODUCED
`11_challenge_funnel.sql`	✅	1 row — the whole-cohort funnel counts
`12_challenge338_user_table.sql`	✅	**1,111 rows** — one per challenge-338 starter
`13_comparison_groups.sql`	✅	**9,956 rows** — one per paying subscriber. The analysis dataset
`14_unsub_timing.sql`	not run	would make unsub % at 12h/24h reportable

The three numbers most likely to be challenged

CLAIM	WHERE TO FIND IT	TEST
The Challenge moved D1 by **+0.5 pts, p = 0.82**	`02_main-output.txt` §4 Test A	Two-proportion z-test, takers (n=1,111) vs exposure-matched control (n=994)
The effect is **not** an activity artefact	`04_loose_ends-output.txt`	Stratified by product activity — flat in 4 of 5 strata
27% of finishers completed it in one sitting	`02_main-output.txt` §5	13 of 48 finishers have `active_challenge_days = 1`

Method notes

- **Confidence intervals** are Wilson score intervals, not normal-approximation — several cells are small and several proportions sit near 0 or 1, where the normal approximation returns impossible bounds.
- **Group comparisons** are two-proportion z-tests with a pooled-variance standard error, reported as difference, z and p rather than as a bare "significant".
- **Censoring** is handled with explicit restricted denominators, stated on every table. The observation window is 14 days (2026-06-12 → 2026-06-25), so D7 is only observable for 72.5% of the cohort and **every unsubscribe rate is a floor**.
- **No regression, no model.** With no hold-out group in the data, the honest tool is a natural control — people who reached the same surface and declined it. Simpler than a model, and it does not hide its assumptions inside coefficients.
- The analysis scripts are **pure Python standard library** (no pandas or numpy on the machine they were written on), so they run anywhere with Python 3 and the two CSVs.

What this appendix cannot tell you

Whether the Challenge was *causal*. There is no hold-out group and no pre/post period anywhere in the warehouse — the release shipped to 100%. Every causal statement in the submission is an observational correction that worked only because a large "looked at it and left" group happened to exist as a natural control. Fixing that is the first recommendation in Part C, not the last.

Verification run — 91 assertions, all passing

```
== cohort shape ==
  ok  subscribers in cohort                                  9956.0000  (doc: 9956)
  ok  challenge-338 starters                                 1111.0000  (doc: 1111)
  ok  non-takers                                             8845.0000  (doc: 8845)
  ok  all rows have a subscription                           9956.0000  (doc: 9956)

== exposure ladder (Part C §1) ==
  ok  saw popup                                              3140.0000  (doc: 3140)
  ok  saw popup %                                            0.3154  (doc: 0.315)
  ok  clicked popup                                          2970.0000  (doc: 2970)
  ok  viewed a challenge                                     2109.0000  (doc: 2109)
  ok  started 338 %                                          0.1116  (doc: 0.112)
  ok  never reached a surface                                6716.0000  (doc: 6716)
  ok  never reached a surface %                              0.6746  (doc: 0.675)

== inside the challenge (Part C §2) ==
  ok  completed >= 1 lessons                                 612.0000  (doc: 612)
  ok  completed >= 2 lessons                                 308.0000  (doc: 308)
  ok  completed >= 3 lessons                                 158.0000  (doc: 158)
  ok  completed >= 8 lessons                                 47.0000  (doc: 47)
  ok  zero lessons %                                         0.4491  (doc: 0.449)
  ok  lesson1->2 survival                                    0.5033  (doc: 0.503)
  ok  completed the course                                   48.0000  (doc: 48)
  ok  certificate                                            23.0000  (doc: 23)

== tiers + headline table (Part C §4) ==
  ok  HIGH n                                                 158.0000  (doc: 158)
  ok  LOW n                                                  953.0000  (doc: 953)
  ok  HIGH D1                                                0.6962  (doc: 0.696)
  ok  LOW D1                                                 0.5845  (doc: 0.584)
  ok  DIDNT-TAKE D1                                          0.2174  (doc: 0.217)
  ok  HIGH D3                                                0.5696  (doc: 0.57)
  ok  LOW D3                                                 0.3442  (doc: 0.344)
  ok  DIDNT-TAKE D3                                          0.1001  (doc: 0.1)
  ok  HIGH D7                                                0.3910  (doc: 0.391)
  ok  HIGH D7 denominator                                    133.0000  (doc: 133)
  ok  LOW D7                                                 0.1805  (doc: 0.18)
  ok  LOW D7 denominator                                     737.0000  (doc: 737)
  ok  DIDNT-TAKE D7                                          0.0689  (doc: 0.069)
  ok  DIDNT-TAKE D7 denominator                              6345.0000  (doc: 6345)
  ok  HIGH unsub                                             0.1139  (doc: 0.114)
  ok  LOW unsub                                              0.1144  (doc: 0.114)
  ok  DIDNT-TAKE unsub                                       0.1431  (doc: 0.143)

== the decisive test (Part C §5 Test A) ==
  ok  takers D1                                              0.6004  (doc: 0.6)
  ok  control n                                              994.0000  (doc: 994)
  ok  control D1 (the 59.6%)                                 0.5956  (doc: 0.596)
  ok  D1 effect (pts)                                        0.0048  (doc: 0.005)
  ok  D1 effect p-value                                      0.8231  (doc: 0.823)
  ok  unsub effect (pts)                                     -0.0225  (doc: -0.023)
  ok  unsub effect p-value                                   0.1189  (doc: 0.119)
  ok  joined-never-started D1                                0.6043  (doc: 0.604)
  ok  viewed-never-joined D1                                 0.5929  (doc: 0.593)

== immortal time + dose-response (Test B / C) ==
  ok  start on day 0 %                                       0.2394  (doc: 0.239)
  ok  day-0 dose effect (pts, negative)                      -0.0920  (doc: -0.092)
  ok  day-0 dose p-value                                     0.0475  (doc: 0.048)

== the mechanic (Test D) ==
  ok  finished in 1 day                                      13.0000  (doc: 13)
  ok  finished in 1 day %                                    0.2708  (doc: 0.271)
  ok  finished in <=2 days %                                 0.4583  (doc: 0.458)
```

```
  ok  returned for a 2nd challenge day %                          0.3676  (doc: 0.368)
  ok  mean lessons per active challenge day                       1.4608  (doc: 1.46)

== the metric that would have fooled the team (§6) ==
  ok  takers D1, challenge-anchored                               0.3978  (doc: 0.398)
  ok  cohort D1, first-app-day                                    0.2602  (doc: 0.26)

== CSAT inversion (§8) ==
  ok  CSAT never took                                             3.7373  (doc: 3.74)
  ok  CSAT LOW                                                    3.5338  (doc: 3.53)
  ok  CSAT HIGH                                                   3.3332  (doc: 3.33)
  ok  CSAT finishers                                              3.1040  (doc: 3.1)
  ok  CSAT of challenge-338 content                               3.2543  (doc: 3.25)
  ok  same users, all content                                     3.4504  (doc: 3.45)

== product-wide baselines (the thesis number) ==
  ok  never completed a lesson anywhere %                         0.5021  (doc: 0.502)
  ok  cohort unsubscribe %                                        0.1399  (doc: 0.14)

== demographics (Part A + the PM's claim) ==
  ok  male %                                                      0.5974  (doc: 0.597)
  ok  female %                                                    0.3841  (doc: 0.384)
  ok  aged 45+ (clean buckets) %                                  0.5776  (doc: 0.578)
  ok  aged 45+ incl. legacy bucket %                              0.6012  (doc: 0.601)
  ok  men 55+ count                                               2120.0000  (doc: 2120)
  ok  men 55+ %                                                   0.2129  (doc: 0.213)
  ok  men 35-54 %                                                 0.2495  (doc: 0.249)
  ok  unsub, age 18-24                                            0.3111  (doc: 0.311)
  ok  unsub, age 25-34                                            0.1970  (doc: 0.197)
  ok  unsub, age 35-44                                            0.1447  (doc: 0.145)
  ok  unsub, age 45-54                                            0.0953  (doc: 0.095)
  ok  unsub, age 55+                                              0.0969  (doc: 0.097)
  ok  unsub, goal 'Work faster'                                   0.1095  (doc: 0.109)
  ok  unsub, goal 'Feel more confident with AI'                   0.1074  (doc: 0.107)
  ok  unsub, goal 'Get a quick side hustle'                       0.3723  (doc: 0.372)

== plans (Part D validation) ==
  ok  1Week share                                                 0.1018  (doc: 0.102)
  ok  4Week share incl. special                                   0.6459  (doc: 0.646)
  ok  12Week share                                                0.2506  (doc: 0.251)
  ok  1Week unsub                                                 0.3462  (doc: 0.346)
  ok  4Week unsub                                                 0.1219  (doc: 0.122)
  ok  12Week unsub                                                0.1058  (doc: 0.106)
  ok  observed-mix blended net LTV                                126.8996  (doc: 126.9)

== unsubscribe reasons (§8) ==
  ok  total unsubscribed                                          1393.0000  (doc: 1393)
  ok  payment/chargeback/dispute unsubs                           157.0000  (doc: 157)
  ok  … as % of unsubs                                            0.1127  (doc: 0.113)

== event catalogue (the 'not observable here' claim) ==
  ok  event catalogue rows (file integrity)                       67.0000  (doc: 67)
  ok  subscription_renewed events                                 17.0000  (doc: 17)
  ok  … per subscriber                                            0.0017  (doc: 0.0017)

======================================================================
All 91 checks passed — every documented number matches the data.
```

Part C Task 1 — the full release analysis

```
================================================================================================================
§1 · THE EXPOSURE LADDER — of every paying subscriber, who ever reaches the Challenge?
================================================================================================================
  step                                        n   % of cohort   step conv.
----------------------------------------------------------------------------------------------------------------
  Paying subscribers in the cohort       9956       100.0%            —
  … saw the Challenge popup              3140        31.5%        31.5%
  … clicked the popup                    2970        29.8%        94.6%
  … viewed a Challenge page              2109        21.2%        71.0%
  … joined Challenge 338                 1341        13.5%        63.6%
  … STARTED Challenge 338                1111        11.2%        82.8%
  … completed ≥1 challenge lesson         613         6.2%        55.2%
  … completed ≥3 challenge lessons        158         1.6%        25.8%
  … completed the whole challenge          48         0.5%        30.4%
  … took the certificate                   23         0.2%        47.9%

  Where the audience is lost (share of the whole paying cohort):
    never reached any challenge surface :   6716  (67.5%)
    reached a surface, never started 338:   2129  (21.4%)
    started 338                         :   1111  (11.2%)

================================================================================================================
§2 · ENGAGEMENT TIERS — definition and calibration
================================================================================================================
  The Challenge ships 8 lessons (confirmed: every user flagged `course_completed`
  has exactly 8 completions). So 'finished' = 8, and the day-3 content sits at lesson 3.

  Tier rule (stated up front, applied everywhere below):
    HIGH        started 338 AND completed >= 3 of the 8 lessons  (got past the Day-2 cliff)
    LOW         started 338 AND completed <= 2 lessons
    DIDN'T TAKE never started 338
  The cut is at 3 because that is where the D7 curve steps (18% -> 39%) and it matches the
  product's own claim ('one day = one skill' -> 3 lessons = a 3-day habit). §4 tests it for
  robustness at every other threshold.

  Challenge-lesson completion histogram (n=1111 starters):
    0 lessons  499   44.9%  cum  44.9%  ████████████████████████
    1 lessons  304   27.4%  cum  72.3%  ███████████████
    2 lessons  150   13.5%  cum  85.8%  ███████
    3 lessons   53    4.8%  cum  90.5%  ███
    4 lessons   25    2.3%  cum  92.8%  ▉
    5 lessons   18    1.6%  cum  94.4%  ▉
    6 lessons   10    0.9%  cum  95.3%  ▉
    7 lessons    5    0.5%  cum  95.8%
    8 lessons   47    4.2%  cum 100.0%  ███

  Lesson-by-lesson survival INSIDE the challenge (of the 1,111 who pressed start):
    reached lesson          n   % of starters   survived prev. step
----------------------------------------------------------------------------------------------------------------
    completed >= 1         612           55.1%                55.1%
    completed >= 2         308           27.7%                50.3%
    completed >= 3         158           14.2%                51.3%
    completed >= 4         105            9.5%                66.5%
    completed >= 5          80            7.2%                76.2%
    completed >= 6          62            5.6%                77.5%
    completed >= 7          52            4.7%                83.9%
    completed >= 8          47            4.2%                90.4%

  Tier sizes:
    HIGH (3+ lessons)         158  (1.6% of subscribers)
    LOW (0–2 lessons)         953  (9.6% of subscribers)
    DIDN'T TAKE IT           8845  (88.8% of subscribers)

================================================================================================================
```

```
§3 · THE HEADLINE COMPARISON  (retention anchored on each user's FIRST APP DAY for all three groups,
     so the groups are on the same clock — see §4 for why this matters)
=================================================================================================
  metric                              HIGH (3+ lessons)         LOW (0–2 lessons)            DIDN'T TAKE IT
-------------------------------------------------------------------------------------------------
  D1 retention  ★TARGET             69.6% [62.1%–76.3%]       58.4% [55.3%–61.5%]       21.7% [20.9%–22.6%]
  D3 retention                      57.0% [49.2%–64.4%]       34.4% [31.5%–37.5%]       10.0% [9.4%–10.6%]
  D7 retention                      39.1% [31.2%–47.6%]       18.0% [15.4%–21.0%]        6.9% [6.3%–7.5%]
  Unsubscribe rate                  11.4% [7.3%–17.3%]        11.4% [9.6%–13.6%]       14.3% [13.6%–15.1%]
    (D7 denominator)                              133                       737                      6345
-------------------------------------------------------------------------------------------------
  Avg lessons completed (all)                   19.28                     6.38                     1.81
  Avg active days in product                     5.37                     3.46                     1.76
  Avg CSAT (1–5)                                 3.33                     3.53                     3.74
    (CSAT raters)                                 158                      845                     3740

  Significance vs 'didn't take it':
    HIGH (3+ lessons)    D1     diff  +47.9%  z=+14.27  p=0.00e+00  ***
    HIGH (3+ lessons)    D3     diff  +47.0%  z=+18.83  p=0.00e+00  ***
    HIGH (3+ lessons)    D7     diff  +32.2%  z=+13.92  p=0.00e+00  ***
    HIGH (3+ lessons)    unsub  diff   -2.9%  z= -1.04  p=2.98e-01  ns
    LOW (0–2 lessons)    D1     diff  +36.7%  z=+24.76  p=0.00e+00  ***
    LOW (0–2 lessons)    D3     diff  +24.4%  z=+21.74  p=0.00e+00  ***
    LOW (0–2 lessons)    D7     diff  +11.2%  z=+10.54  p=0.00e+00  ***
    LOW (0–2 lessons)    unsub  diff   -2.9%  z= -2.43  p=1.52e-02  *

=================================================================================================
§4 · IS THE GAP CAUSAL? — three tests
=================================================================================================
  TEST A — EXPOSURE-MATCHED CONTROL.
  'Didn't take it' is not one group: most of them were never offered the Challenge.
  The fair control is people who DID reach the Challenge and chose not to engage.

  exposure group                        n                   D1                    D3               unsub
-------------------------------------------------------------------------------------------------
  started 338 (takers)               1111   60.0% [57.1%–62.9%]   37.6% [34.8%–40.5%]  11.4% [9.7%–13.4%]
  joined 338, never started           230   60.4% [54.0%–66.5%]   37.8% [31.8%–44.2%] 13.9% [10.0%–19.0%]
  viewed 338, never joined            764   59.3% [55.8%–62.7%]   36.6% [33.3%–40.1%] 13.6% [11.4%–16.2%]
  saw popup, never opened 338        1132   44.9% [42.0%–47.8%]   24.0% [21.6%–26.6%] 25.5% [23.1%–28.1%]
  never reached a surface            6716   12.2% [11.4%–13.0%]      3.7% [3.2%–4.1%] 12.5% [11.7%–13.3%]

  takers vs exposed-but-didn't-start:  D1 diff +0.5%  z=+0.22  p=0.823  ns
  takers vs exposed-but-didn't-start:  unsub diff -2.3%  z=-1.56  p=0.119  ns


  TEST B — IMMORTAL-TIME BIAS.
  Only 24% of takers start on their first app day; the median taker starts on day +1.
  To start on day +k you must already have survived to day k — so takers are pre-selected
  survivors. The clean read is the 'day-0 starters' who had no survival head start.

  taker start-day lag: day0 266 (24%), day+1 342 (31%), day+2 196 (18%), day+3 or later 307 (28%)

  group                                       n                   D1                    D3
-------------------------------------------------------------------------------------------------
  takers who started on day 0               266   39.1% [33.4%–45.1%]   22.6% [17.9%–27.9%]
  all takers (any start day)               1111   60.0% [57.1%–62.9%]   37.6% [34.8%–40.5%]
  never took it                            8845   21.7% [20.9%–22.6%]    10.0% [9.4%–10.6%]

  day-0 takers vs never-took:  D1 diff +17.4%  z=+6.71  p=2.00e-11  ***


  TEST C — A NON-TAUTOLOGICAL DOSE-RESPONSE.
  'More lessons -> better retention' is partly circular: doing lesson 3 on day 3 IS retention.
  The clean version: among users whose challenge activity happened on ONE day only, does a
  bigger DAY-0 dose predict coming back tomorrow? Nothing circular here.

  n = 816 takers with all challenge activity on a single day
  day-0 lessons         n             D1 next day               unsub
-------------------------------------------------------------------------------------------------
```

```
  0                   429     55.7% [51.0%–60.3%]  10.5% [7.9%–13.7%]
  1                   248     54.4% [48.2%–60.5%]  12.5% [8.9%–17.2%]
  2                    99     50.5% [40.8%–60.1%]  11.1% [6.3%–18.8%]
  3                    21     33.3% [17.2%–54.6%] 23.8% [10.6%–45.1%]
  4                     5       0.0% [0.0%–43.4%]  20.0% [3.6%–62.4%]
  8                    13     53.8% [29.1%–76.8%] 38.5% [17.7%–64.5%]

  >=2 vs <=1 day-0 lessons:  D1 diff -9.2%  z=-1.98  p=0.048  *


  THRESHOLD ROBUSTNESS — the 'high' cut at every possible value:
  high = >= k lessons     n high     D1 high      D1 low    D1 gap   unsub high   unsub low
-----------------------------------------------------------------------------------------------
  1                         612       61.4%       58.3%     +3.1%        12.7%        9.8%
  2                         308       63.3%       58.8%     +4.5%        13.0%       10.8%
  3                         158       69.6%       58.4%    +11.2%        11.4%       11.4%
  4                         105       74.3%       58.5%    +15.7%        10.5%       11.5%
  5                          80       78.8%       58.6%    +20.2%        10.0%       11.5%
  6                          62       80.6%       58.8%    +21.8%        11.3%       11.4%
  7                          52       76.9%       59.2%    +17.7%        11.5%       11.4%
  8                          47       76.6%       59.3%    +17.3%        12.8%       11.4%

===============================================================================================
§5 · THE MECHANIC CHECK — is this actually a DAILY challenge?
===============================================================================================
  The hypothesis was 'one day = one skill builds a daily habit'. That only works if the
  content is gated to one lesson per day. Test: how many lessons do people do per active day?

  Users who completed the whole 8-lesson challenge: 48 (4.3% of starters, 0.48% of subscribers)

  … and how many DAYS they took to do it:
     1 day(s):  13  ( 27.1%)  cum  27.1%  ████████████
     2 day(s):   9  ( 18.8%)  cum  45.8%  ████████
     3 day(s):   6  ( 12.5%)  cum  58.3%  ██████
     4 day(s):   4  (  8.3%)  cum  66.7%  ████
     5 day(s):   5  ( 10.4%)  cum  77.1%  ████
     6 day(s):   2  (  4.2%)  cum  81.2%  ██
     7 day(s):   3  (  6.2%)  cum  87.5%  ███
     8 day(s):   3  (  6.2%)  cum  93.8%  ███
     9 day(s):   2  (  4.2%)  cum  97.9%  ██
    10 day(s):   1  (  2.1%)  cum 100.0%  █

  → 13/48 (27%) finished the '7-day' challenge in ONE day.
  → 22/48 (46%) finished it in two days or fewer.

  Lessons completed per active challenge day (all takers who did >=1 lesson):
    ~0 lessons/day:   63  (10.3%)
    ~1 lessons/day:  357  (58.3%)
    ~2 lessons/day:  136  (22.2%)
    ~3 lessons/day:   28  (4.6%)
    ~4 lessons/day:   14  (2.3%)
    ~5 lessons/day:    1  (0.2%)
    ~8 lessons/day:   13  (2.1%)

  mean lessons per active challenge day: 1.46

  Return behaviour: of takers who did >=1 lesson, how many ever came back for a 2nd challenge day?
    225/612 = 36.8%

===============================================================================================
§6 · SEGMENTS
===============================================================================================

  —— AGE ——
  segment                              subs  % cohort   take rate    D1 (all)     unsub   D1 takers   D1
non-tk
-----------------------------------------------------------------------------------------------
  55+                                  3200     32.1%       10.2%       25.4%      9.7%       60.9%
21.4%
```

```
  45-54                          2551     25.6%       10.9%       26.9%      9.5%       62.1%
22.6%
  35-44                          1887     19.0%       12.0%       27.8%     14.5%       66.1%
22.5%
  25-34                          1117     11.2%       12.4%       24.4%     19.7%       54.0%
20.1%
  18-24                           913      9.2%       10.3%       21.9%     31.1%       44.7%
19.3%
  45+                             235      2.4%       20.0%       37.0%     17.0%       59.6%
31.4%
  (unknown)                        53      0.5%        3.8%       13.2%     43.4%      100.0%
9.8%

  ── GENDER ──
  segment                        subs  % cohort   take rate    D1 (all)     unsub   D1 takers   D1
non-tk
-------------------------------------------------------------------------------------------------
  Male                           5948     59.7%       11.3%       27.4%     12.5%       62.3%
23.0%
  Female                         3824     38.4%       11.0%       23.8%     16.1%       56.2%
19.8%
  I'd rather skip this one        122      1.2%       10.7%       31.1%      6.6%       69.2%
26.6%
  (unknown)                        62      0.6%        4.8%       16.1%     38.7%       66.7%
13.6%

  ── PLAN ──
  segment                        subs  % cohort   take rate    D1 (all)     unsub   D1 takers   D1
non-tk
-------------------------------------------------------------------------------------------------
  4Week                          6341     63.7%       11.2%       26.5%     12.2%       60.5%
22.2%
  12Week                         2495     25.1%       11.7%       26.4%     10.6%       61.3%
21.8%
  1Week                          1014     10.2%        9.5%       20.9%     34.6%       51.0%
17.8%
  4Week_special                    90      0.9%       12.2%       33.3%      3.3%       72.7%
27.8%

  ── WORK STATUS ──
  segment                        subs  % cohort   take rate    D1 (all)     unsub   D1 takers   D1
non-tk
-------------------------------------------------------------------------------------------------
  Full-time employee             3341     33.6%       11.6%       27.1%     11.8%       61.2%
22.6%
  Business owner                 1817     18.3%        8.8%       24.9%      9.4%       59.4%
21.5%
  Freelancer / Self-employed     1487     14.9%       11.6%       26.1%     10.5%       63.0%
21.2%
  Exploring options               845      8.5%       12.3%       26.0%     18.6%       55.8%
21.9%
  Full-time worker                731      7.3%       12.3%       28.5%     28.5%       57.8%
24.3%
  Between jobs / Career switcher  578      5.8%       13.1%       26.5%     13.8%       61.8%
21.1%
  Student                         175      1.8%       14.9%       20.0%     37.1%       53.8%
14.1%

  ── GOAL ──
  segment                        subs  % cohort   take rate    D1 (all)     unsub   D1 takers   D1
non-tk
-------------------------------------------------------------------------------------------------
  Work faster                    3015     30.3%       10.0%       25.4%     10.9%       60.4%
21.5%
  Feel more confident with AI    1992     20.0%       10.3%       25.5%     10.7%       62.6%
21.2%
  Start my own business          1161     11.7%       14.4%       29.8%     10.9%       64.7%
23.9%
  Earn more                       830      8.3%       11.3%       27.7%     10.7%       64.9%
```

```
23.0%
  Get a promotion or a better job       636      6.4%       11.0%       25.0%     14.0%       50.0%
21.9%
  Gain flexibility or work remotely     492      4.9%       13.4%       25.2%     27.4%       57.6%
20.2%
  Get a quick side hustle               368      3.7%       10.6%       21.2%     37.2%       51.3%
17.6%
  Unlock better income opportunitie     347      3.5%       15.3%       31.1%     25.4%       50.9%
27.6%
  Increase work productivity            185      1.9%       10.8%       24.3%     10.3%       60.0%
20.0%

  ── COUNTRY ──
  segment                              subs  % cohort   take rate    D1 (all)     unsub   D1 takers   D1
non-tk
-------------------------------------------------------------------------------------------------------
  US                                   3749     37.7%        9.8%       24.2%     12.5%       60.1%
20.3%
  GB                                   1020     10.2%       10.0%       26.6%     12.6%       65.7%
22.2%
  AU                                    691      6.9%       10.9%       25.6%     16.4%       60.0%
21.4%
  CA                                    523      5.3%       12.4%       24.5%     16.3%       55.4%
20.1%
  IT                                    375      3.8%       14.4%       26.1%     17.3%       53.7%
21.5%
  AE                                    263      2.6%       14.8%       26.6%     12.2%       41.0%
24.1%
  DE                                    255      2.6%       11.4%       26.7%     20.8%       79.3%
19.9%
  FR                                    250      2.5%       10.4%       23.6%     14.4%       57.7%
19.6%
  SG                                    243      2.4%       12.3%       30.5%     10.7%       56.7%
26.8%
  ES                                    229      2.3%       14.8%       24.5%     14.0%       52.9%
19.5%
  MX                                    202      2.0%        8.4%       29.2%      5.4%       88.2%
23.8%
  CO                                    154      1.5%       10.4%       24.0%      9.1%       56.2%
20.3%

  ── AGE × GENDER (the PM's claim: 'our main market is 40–50 y.o. men') ──
  age               Male      Female    other/NA   row total    row %
-------------------------------------------------------------------------------------------------------
  18-24              602         310           1         913     9.2%
  25-34              613         493          11        1117    11.2%
  35-44             1040         815          32        1887    19.0%
  45-54             1444        1065          42        2551    25.6%
  55+               2120        1035          45        3200    32.1%
  TOTAL             5819        3718         131        9956
                  58.4%       37.3%

  Men aged 35–54            :   2484  (24.9% of paying subscribers)
  Everyone aged 35–54       :   4438  (44.6%)
  All men                   :   5819  (58.4%)
  All women                 :   3718  (37.3%)
  Aged 45+                  :   5751 (57.8%)

=======================================================================================================
§7 · UNSUBSCRIBES
=======================================================================================================
  total unsubscribed in cohort: 1393 / 9956 = 14.0%

  plan                subs   unsub                  rate
-------------------------------------------------------------------------------------------------------
  4Week               6341     773   12.2% [11.4%–13.0%]
  12Week              2495     264    10.6% [9.4%–11.8%]
  1Week               1014     351   34.6% [31.8%–37.6%]
  4Week_special         90       3      3.3% [1.1%–9.3%]
```

```
  tier                          n   unsub                rate
-------------------------------------------------------------------------------------------------
  HIGH (3+ lessons)           158      18    11.4% [7.3%–17.3%]
  LOW (0–2 lessons)           953     109    11.4% [9.6%–13.6%]
  DIDN'T TAKE IT             8845    1266   14.3% [13.6%–15.1%]

  Stated reason (only recorded for some unsubscribes):
    (none recorded)               873  (62.7% of unsubs)
    Support request               310  (22.3% of unsubs)
    hard_decline                   84  (6.0% of unsubs)
    account_deletion               31  (2.2% of unsubs)
    Mastercard Alert               30  (2.2% of unsubs)
    general                        22  (1.6% of unsubs)
    dispute                        12  (0.9% of unsubs)
    paypal_refund                  10  (0.7% of unsubs)
    Visa CDRN                       7  (0.5% of unsubs)
    fraud                           6  (0.4% of unsubs)
    Merchanto visa                  4  (0.3% of unsubs)
    payment_refunded                3  (0.2% of unsubs)
    fraud_reported                  1  (0.1% of unsubs)

  → payment-failure / chargeback / refund-dispute reasons: 157 (11.3% of unsubs, 1.6% of the cohort)

  Do challenge takers unsubscribe less? (raw, uncontrolled)
    takers           127/1111   = 11.4% [9.7%–13.4%]
    non-takers      1266/8845   = 14.3% [13.6%–15.1%]
    diff -2.9%  z=-2.61  p=0.009  **
```

Decisions settled before the tiers were chosen

```
subscribers: 9956   ·   challenge-338 takers: 1111

================================================================================
Q1 · lag between first app day and challenge start (days)
================================================================================
  + 0 days:   266  ( 23.9%)  cum  23.9%
  + 1 days:   342  ( 30.8%)  cum  54.7%
  + 2 days:   196  ( 17.6%)  cum  72.4%
  + 3 days:    90  (  8.1%)  cum  80.5%
  + 4 days:    65  (  5.9%)  cum  86.3%
  + 5 days:    58  (  5.2%)  cum  91.5%
  + 6 days:    42  (  3.8%)  cum  95.3%
  + 7 days:    24  (  2.2%)  cum  97.5%
  + 8 days:    15  (  1.4%)  cum  98.8%
  + 9 days:     3  (  0.3%)  cum  99.1%
  +10 days:     6  (  0.5%)  cum  99.6%
  +11 days:     2  (  0.2%)  cum  99.8%
  +12 days:     1  (  0.1%)  cum  99.9%
  +13 days:     1  (  0.1%)  cum 100.0%

  → Only 23.9% of takers start the challenge on their FIRST app day;
    the median taker starts on day +1, and 28% start on day +3 or later.
    CONSEQUENCE: the two anchors are NOT interchangeable, and takers are pre-selected
    survivors — to start on day k you must have survived to day k (immortal-time bias).
    So: use the first-seen anchor for all cross-group comparisons, and treat even that
    as favourable to takers. See 02_main.py §4 Test B.

================================================================================
Q2 · challenge lessons STARTED vs COMPLETED (the >7 spike)
================================================================================
  completed: {0: 499, 1: 304, 2: 150, 3: 53, 4: 25, 5: 18, 6: 10, 7: 5, 8: 47}
  started:   {0: 1, 1: 698, 2: 234, 3: 60, 4: 32, 5: 19, 6: 12, 7: 6, 8: 49}

  users with >7 completions: 47
  their completed-course flag: Counter({1: 47})
  their certificate flag:      Counter({0: 25, 1: 22})
  their active_challenge_days: {1: 13, 2: 9, 3: 5, 4: 4, 5: 5, 6: 2, 7: 3, 8: 3, 9: 2, 10: 1}
  their lessons_started:       {8: 47}

  cross-check — completed_course flag by completion count:
    0 lessons: course_completed    0/ 499
    1 lessons: course_completed    0/ 304
    2 lessons: course_completed    0/ 150
    3 lessons: course_completed    0/  53
    4 lessons: course_completed    0/  25
    5 lessons: course_completed    0/  18
    6 lessons: course_completed    0/  10
    7 lessons: course_completed    1/   5
    8 lessons: course_completed   47/  47

================================================================================
Q3 · DOSE-RESPONSE: outcome by exact number of challenge lessons completed
     (retention anchored on first_seen so it is comparable across everyone)
================================================================================
   lessons     n |                  D1                  D3                 D7 (n obs) |              unsub
  ---------------------------------------------------------------------------------------------------------
         0   499 |  58.3% [53.9%–62.6%]  33.5% [29.5%–37.7%]  18.5% [15.0%–22.7%] (383) |  9.8% [7.5%–12.7%]
         1   304 |  59.5% [53.9%–64.9%]  34.5% [29.4%–40.0%]  17.2% [13.0%–22.5%] (238) | 12.5% [9.2%–16.7%]
         2   150 |  56.7% [48.7%–64.3%]  37.3% [30.0%–45.3%]  18.1% [12.2%–26.1%] (116) | 14.7% [9.9%–21.2%]
         3    53 |  60.4% [46.9%–72.4%]  35.8% [24.3%–49.3%]  39.0% [25.7%–54.3%] ( 41) | 13.2% [6.5%–24.8%]
         4    25 |  60.0% [40.7%–76.6%]  68.0% [48.4%–82.8%]  31.8% [16.4%–52.7%] ( 22) | 12.0% [4.2%–30.0%]
         5    18 |  72.2% [49.1%–87.5%]  61.1% [38.6%–79.7%]  38.5% [17.7%–64.5%] ( 13) |  5.6% [1.0%–25.8%]
         6    10 | 100.0% [72.2%–100.0%]  80.0% [49.0%–94.3%]  50.0% [21.5%–78.5%] (  8) | 10.0% [1.8%–
40.4%]
```

```
         7     5 |  80.0% [37.6%–96.4%] 100.0% [56.6%–100.0%]  40.0% [11.8%–76.9%] (  5) |  0.0% [0.0%–
43.4%]
         8    47 |  76.6% [62.8%–86.4%]  63.8% [49.5%–76.0%]  40.9% [27.7%–55.6%] ( 44) | 12.8% [6.0%–25.2%]

  same, by ACTIVE CHALLENGE DAYS (the mechanic's own unit: 1 day = 1 skill)
   lessons     n |                   D1                   D3                 D7 (n obs) |              unsub
  ------------------------------------------------------------------------------------------------------------
         1   816 |  53.7% [50.2%–57.1%]  29.4% [26.4%–32.6%]  14.7% [12.1%–17.7%] (614) | 12.0% [10.0%–
14.4%]
         2   163 |  76.1% [69.0%–82.0%]  46.6% [39.1%–54.3%]  28.6% [21.7%–36.5%] (140) | 11.0% [7.1%–16.8%]
         3    59 |  76.3% [64.0%–85.3%]  74.6% [62.2%–83.9%]  36.0% [24.1%–49.9%] ( 50) |  8.5% [3.7%–18.4%]
         4    29 |  79.3% [61.6%–90.2%]  79.3% [61.6%–90.2%]  39.1% [22.2%–59.2%] ( 23) | 10.3% [3.6%–26.4%]
         5    21 |  90.5% [71.1%–97.3%]  76.2% [54.9%–89.4%]  52.4% [32.4%–71.7%] ( 21) | 14.3% [5.0%–34.6%]
         6     8 |  62.5% [30.6%–86.3%] 100.0% [67.6%–100.0%]  71.4% [35.9%–91.8%] (  7) |  0.0% [0.0%–
32.4%]
         7    15 |  86.7% [62.1%–96.3%]  73.3% [48.0%–89.1%]  80.0% [54.8%–93.0%] ( 15) |  0.0% [0.0%–20.4%]

================================================================================
Q4 · observation window / censoring
================================================================================
  last data date: 2026-06-25
  days from first_seen to last data date:
     0 days:     3
     1 days:    10
     2 days:     2
     3 days:    10
     4 days:   862
     5 days:   878
     6 days:   976
     7 days:  1167
     8 days:  1270
     9 days:  1051
    10 days:   911
    11 days:  1057
    12 days:   877
    13 days:   882
  D1 observable for 9953/9956 (100.0%) of subscribers
  D3 observable for 9941/9956 (99.8%) of subscribers
  D7 observable for 7215/9956 (72.5%) of subscribers

================================================================================
Q5 · is challenge 338 effectively THE Challenge feature?
================================================================================
  started_any_challenge: 1114
  group_338 == took_338: 1111
  took_other_challenge : 3
  → 338 covers 99.7% of all challenge starters.
```

Supplementary tests — CSAT, refund gate, data quality

```
================================================================================================
§A · D1 AS THE TEAM DEFINED IT vs D1 ON A COMMON CLOCK
================================================================================================
  Brief: 'Target metric: D1 Retention (retention the day after the challenge started).'
  That anchors takers on their challenge-start day. Non-takers have no such day, so the only
  way to compare is a common clock (first app day). Both are shown — the difference is the
  single biggest reason a naive read of this release looks better than it is.

  measure                                                    n                    rate
------------------------------------------------------------------------------------------------
  TAKERS · D1 anchored on challenge-start (team defn)    1111     39.8% [36.9%–42.7%]
  TAKERS · D1 anchored on first app day                  1111     60.0% [57.1%–62.9%]
  NON-TAKERS · D1 anchored on first app day              8845     21.7% [20.9%–22.6%]
  TAKERS · D3 anchored on challenge-start                 982     25.9% [23.2%–28.7%]
  TAKERS · D7 anchored on challenge-start                 606     14.7% [12.1%–17.7%]

================================================================================================
§B · CSAT — does engaging with the product make people happier?
================================================================================================
  group                                       raters   mean CSAT     % 4-5     % 1-2
------------------------------------------------------------------------------------------------
  Completed the whole challenge                 48        3.10     37.5%     22.9%
  HIGH  (3+ challenge lessons)                 158        3.33     38.6%     19.6%
  LOW   (0-2 challenge lessons)                845        3.53     53.0%     20.6%
  Never took the challenge                    3740        3.74     59.8%     17.5%

  CSAT given INSIDE challenge 338 specifically: n=583 raters, mean 3.25
  … of 1111 starters, only 52.5% ever rated challenge content.
  Same users' CSAT across ALL product content: 3.45

================================================================================================
§C · COMPLETION AND THE REFUND GATE
================================================================================================
  Part A found (from public reviews) that finishing the plan is the money-back gate.
  If true, completion should predict MORE unsubscribing, not less. Test it.

  group                                              n          unsubscribe rate
------------------------------------------------------------------------------------------------
  finished the challenge (all 8 lessons)            48        12.5% [5.9%–24.7%]
  started but did not finish                      1063        11.4% [9.6%–13.4%]
  never took it                                   8845       14.3% [13.6%–15.1%]

  finishers vs non-finishers: diff +1.1%  z=+0.24  p=0.812  ns

  of the finishers, those who binged it in <=2 days: n=22, unsub 22.7% [10.1%–43.4%]
  of the finishers, those who spread it over 3+ days: n=26, unsub 3.8% [0.7%–18.9%]

  took the certificate: n=23, unsub 13.0% [4.5%–32.1%]

================================================================================================
§D · THE 'SAW THE POPUP AND WALKED AWAY' GROUP — the worst cell in the dataset
================================================================================================
  group                                          n                  D1               unsub   avg lessons
------------------------------------------------------------------------------------------------
  saw popup, never opened the challenge      1132 44.9% [42.0%–47.8%] 25.5% [23.1%–28.1%]          3.71
  never reached any challenge surface        6716 12.2% [11.4%–13.0%] 12.5% [11.7%–13.3%]          0.81
  took the challenge                         1111 60.0% [57.1%–62.9%]  11.4% [9.7%–13.4%]          8.22

  plan mix of 'saw popup, never opened': [('4Week', 703), ('12Week', 266), ('1Week', 144), ('4Week_special',
15)]
  plan mix of the whole cohort         : [('4Week', 6341), ('12Week', 2495), ('1Week', 1014),
('4Week_special', 90)]
```

```
  1-Week subscribers as share of 'saw popup, never opened': 12.7%  (cohort: 10.2%)

================================================================================================
§E · DATA-QUALITY FLAGS (things I would raise before anyone acts on this)
================================================================================================
  1. Two different quiz vocabularies are mixed in one cohort:
     age      '45-54': n=2551  unsub=  9.5% D1= 26.9%
     age      '55+': n=3200  unsub=  9.7% D1= 25.4%
     age      '45+': n=235   unsub= 17.0% D1= 37.0%
     status   'Full-time employee': n=3341  unsub= 11.8% D1= 27.1%
     status   'Full-time worker': n=731   unsub= 28.5% D1= 28.5%
     → 'Full-time worker' churns at 2.4x 'Full-time employee'. Same words, different funnel.
       Either a different quiz version or different traffic. Cannot be pooled blindly.

  2. THE BIG ONE — there is no hold-out group. Challenge starts run on every single day of
     the window (2026-06-12 … 2026-06-25), so there is no pre/post and no control arm.
     Every causal statement below is an observational correction, not an experiment.
     This is the #1 thing to fix before the next release, not after it.

  3. Observation window is only 14 days (2026-06-12 → 2026-06-25):
     D7 is unobservable for 27.5% of the cohort;
     unsubscribe is right-censored for everyone (a 4-week plan cannot even reach its
     first renewal inside this window), so all unsubscribe rates here are FLOOR values.

  4. The popup carries no challenge_id, so 'saw the popup' is exposure to ANY challenge,
     not specifically 338. 338 is 99.7% of all starts, so the practical impact is small.

================================================================================================
§F · WHAT THE FEATURE IS WORTH TODAY
================================================================================================
  Reach:        1111/9956 = 11.2% of paying subscribers started it.
  Activation:   612/1111 = 55.1% did even one lesson.
  Completion:   48/1111 = 4.3% finished (0.48% of all subscribers).

  Best causal estimate of the D1 effect (vs the exposure-matched control):
    +0.5% percentage points, p=0.823 (ns) — statistically indistinguishable from zero.

  Even if the effect were real and equal to the raw taker-vs-non-taker gap, at 11.2% reach
  the cohort-level D1 impact would be at most 0.112 x the gap.
    raw gap = +38.3%  →  cohort-level ceiling = +4.3% pts of overall D1.
  Overall cohort D1 today: 26.0%
```

Cohort baselines and the stratified control test

```
== cohort-level baselines (all 9,956 paying subscribers, first-app-day anchor) ==
  D1: 26.0% [25.2%–26.9%]   (n=9953)
  D3: 13.1% [12.5%–13.8%]   (n=9941)
  D7: 8.6% [8.0%–9.3%]   (n=7215)
  unsubscribed within the 14-day window: 14.0% [13.3%–14.7%]
  mean lessons completed (any course): 2.53
  mean active days in product: 1.98
  never completed a single lesson anywhere: 50.2%
  finished onboarding: 69.8%

== CONFOUND CHECK for §D: is 'saw the popup' just a proxy for time-in-product? ==
  group                            n  mean active days  mean days observed     unsub
  took 338                      1111              3.73                8.94     11.4%
  saw popup, never opened       1132              2.74                8.95     25.5%
  never reached a surface       6716              1.33                8.27     12.5%

  Same three groups, holding ACTIVE DAYS roughly constant (users with 3-6 active days):
  group                            n                  D1               unsub
  took 338                       622 69.6% [65.9%–73.1%]  11.3% [9.0%–14.0%]
  saw popup, never opened        516 63.4% [59.1%–67.4%] 24.0% [20.5%–27.9%]
  never reached a surface        326 58.9% [53.5%–64.1%] 19.3% [15.4%–24.0%]

== TAKERS vs EXPOSED-CONTROL, holding active days constant (the tightest test I can run) ==
  active days     n takers     D1 takers  n control    D1 control     diff        p
  1                     97          0.0%         63          0.0%    +0.0%    1.000
  2                    270         46.3%        257         42.4%    +3.9%    0.370
  3-4                  446         64.8%        427         65.1%    -0.3%    0.924
  5-7                  217         82.9%        205         84.4%    -1.4%    0.689
  8+                    81         90.1%         42         76.2%   +13.9%    0.038

== was the Challenge live for the whole observation window? ==
  2026-06-12:   39 starts
  2026-06-13:   69 starts
  2026-06-14:  101 starts
  2026-06-15:  124 starts
  2026-06-16:   70 starts
  2026-06-17:   88 starts
  2026-06-18:  115 starts
  2026-06-19:  102 starts
  2026-06-20:   84 starts
  2026-06-21:   84 starts
  2026-06-22:  106 starts
  2026-06-23:   76 starts
  2026-06-24:   38 starts
  2026-06-25:   15 starts

== plan mix: real cohort vs the plans.csv assumption used in Part D ==
  1Week :  10.2%   (plans.csv assumes 10%)
  4Week :  64.6%   (plans.csv assumes 70%)
  12Week:  25.1%   (plans.csv assumes 20%)
```

Part A — k-modes segment clustering

```
quiz respondents in export        49,487
answered >= 6 of 10 core questions  38,071  (76.9%)  <- the clustering sample

== choosing k: within-cluster Hamming cost ==
   k     cost   cost/user   improvement
   2  193,760      5.089       —
   3  182,252      4.787     5.9%
   4  177,769      4.669     2.5%
   5  171,100      4.494     3.8%
   6  167,414      4.397     2.2%
   7  164,227      4.314     1.9%
   8  159,839      4.198     2.7%

== k = 6 · cluster profiles (lift = share in cluster / share in population) ==

  ── cluster 0  n = 10,810  (28.4%)
      1.87x  62.3%  feels_ready      Somewhat ready — but I could be behind
      1.63x  49.6%  learn_claude_for Growth - I love learning in-demand skills
      1.61x  42.7%  benefit          Feel more confident with AI
      1.52x  41.8%  age              45-54
      1.33x  69.7%  ai_experience    Good, but i feel like i’m only scratching the surfac
      1.28x  75.0%  gender           Male

  ── cluster 3  n = 8,372  (22.0%)
      2.22x  50.3%  feels_ready      A bit worried — things are changing quickly
      2.17x  61.0%  career_fear      Falling behind as others move faster
      1.87x  82.0%  used_claude      Not yet
      1.59x  40.4%  field            Other
      1.43x  48.9%  age              55+
      1.29x  44.6%  benefit          Work faster

  ── cluster 1  n = 7,953  (20.9%)
      2.51x  45.7%  feels_ready      Yes - I’m confident in my skills
      2.50x  60.6%  ai_experience    Great - AI already helps me a lot
      1.71x  67.2%  gender           Female
      1.66x  57.1%  benefit          Work faster
      1.49x  80.1%  learn_claude_for Work tasks
      1.39x  58.2%  career_fear      Nothing - I see AI as an opportunity, not a threat

  ── cluster 2  n = 4,699  (12.3%)
      2.59x  49.8%  age              35-44
      1.49x  51.4%  benefit          Work faster
      1.39x  75.0%  learn_claude_for Work tasks
      1.31x  76.6%  gender           Male
      1.28x  71.5%  used_claude      Yes
      1.17x  49.6%  work_status      Full-time employee

  ── cluster 4  n = 3,413  (9.0%)
      3.11x  50.4%  benefit          Start my own business
      2.56x  46.5%  feels_ready      Not really — I need new skills fast
      2.16x  84.9%  gender           Female
      2.07x  52.7%  field            Other
      1.99x  60.8%  learn_claude_for Growth - I love learning in-demand skills
      1.77x  48.6%  age              45-54

  ── cluster 5  n = 2,824  (7.4%)
      4.42x  49.6%  work_status      Exploring options
      2.43x  44.2%  feels_ready      Not really — I need new skills fast
      2.32x  27.2%  ai_experience    I haven’t really tried yet
      2.22x  58.6%  benefit          Feel more confident with AI
      2.04x  31.9%  learn_claude_for Personal use
      2.01x  68.9%  age              55+
```

SQL · 00_explore_schema

```
-- Part C · Step 0 — LEARN THE REAL SCHEMA FIRST (run this before anything else)
-- Project: persona-496908 · Tables: app_events, subscribe_events
-- The queries in 01–04 use ASSUMED column names (flagged with -- ⚠ADJUST). Run these
-- three first, then find/replace the assumed names with the real ones.

-- 1) What columns exist? (replace <dataset> with the real dataset id shown in the console)
SELECT table_name, column_name, data_type
FROM `persona-496908.<dataset>.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name IN ('app_events','subscribe_events')
ORDER BY table_name, ordinal_position;

-- 2) What do app_events rows look like, and what event names exist?
SELECT * FROM `persona-496908.<dataset>.app_events` LIMIT 50;

SELECT event_name, COUNT(*) AS n           -- ⚠ADJUST: the event-name column may be `event`, `name`, `type`
FROM `persona-496908.<dataset>.app_events`
GROUP BY event_name ORDER BY n DESC LIMIT 100;
-- ↑ Look here for the Challenge events: something like challenge_start / challenge_day_complete /
--   lesson_complete / day_open. Note the exact strings — 01–04 depend on them.

-- 3) What does subscribe_events carry? (need: user id, subscription start, cancel/unsubscribe, plan, price)
SELECT * FROM `persona-496908.<dataset>.subscribe_events` LIMIT 50;
```

SQL · 01_engagement_tiers

```
-- Part C · Step 1 — Classify each Challenge-eligible user into an engagement tier.
-- ⚠ADJUST all names marked below to the real schema found in 00_explore_schema.sql.
-- Tiers:  high = started AND (reached >=Day3 OR returned on >=3 of first 7 days)
--         low  = started but stalled before Day 3
--         none = eligible but never opened Day 1

DECLARE challenge_id STRING DEFAULT '338';          -- the Challenge in the brief (skills/challenges/338)

WITH ev AS (
  SELECT
    user_id,                                         -- ⚠ADJUST
    event_name,                                      -- ⚠ADJUST
    TIMESTAMP(event_timestamp) AS ts,                -- ⚠ADJUST
    DATE(TIMESTAMP(event_timestamp)) AS d,
    SAFE_CAST(JSON_VALUE(event_params, '$.challenge_id') AS STRING) AS ch  -- ⚠ADJUST (may be a column)
  FROM `persona-496908.<dataset>.app_events`
),
challenge AS (                                       -- events belonging to THIS challenge
  SELECT * FROM ev WHERE ch = challenge_id OR ch IS NULL   -- drop the OR NULL once the field is confirmed
),
per_user AS (
  SELECT
    user_id,
    MIN(IF(event_name='challenge_start', ts, NULL))                    AS started_at,     -- ⚠ADJUST event
    COUNTIF(event_name='challenge_day_complete')                        AS days_completed, -- ⚠ADJUST
event
    COUNT(DISTINCT IF(event_name='challenge_day_complete', d, NULL))    AS distinct_active_days,
    MIN(d) AS first_day, MAX(d) AS last_day
  FROM challenge
  GROUP BY user_id
)
SELECT
  user_id,
  started_at IS NOT NULL AS started,
  days_completed,
  distinct_active_days,
  CASE
    WHEN started_at IS NULL THEN 'none'
    WHEN days_completed >= 3 OR distinct_active_days >= 3 THEN 'high'
    ELSE 'low'
  END AS engagement_tier
FROM per_user;

-- Sanity: the tier distribution (use this histogram to calibrate the Day-3 threshold)
-- SELECT engagement_tier, COUNT(*) FROM ( <the query above> ) GROUP BY 1;
```

SQL · 02_core_metrics

```
-- Part C · Step 2 — The core table: target + secondary metrics BY engagement tier.
-- Depends on the tiering in 01. ⚠ADJUST names to the real schema.
-- D1/D3/D7 retention are measured relative to each user's challenge-start day.

DECLARE challenge_id STRING DEFAULT '338';

WITH tiers AS (
  -- paste/inline the SELECT from 01_engagement_tiers.sql, or save it as a view and reference it
  SELECT * FROM `persona-496908.<dataset>.v_challenge_tiers`     -- ⚠ create this view from 01
),
starts AS (
  SELECT user_id, DATE(MIN(TIMESTAMP(event_timestamp))) AS d0    -- ⚠ADJUST
  FROM `persona-496908.<dataset>.app_events`
  WHERE event_name='challenge_start'                              -- ⚠ADJUST
  GROUP BY user_id
),
activity AS (
  SELECT DISTINCT user_id, DATE(TIMESTAMP(event_timestamp)) AS d  -- any app activity = "retained that day"
  FROM `persona-496908.<dataset>.app_events`
),
ret AS (
  SELECT s.user_id,
    MAX(a.d = DATE_ADD(s.d0, INTERVAL 1 DAY)) AS d1,
    MAX(a.d = DATE_ADD(s.d0, INTERVAL 3 DAY)) AS d3,
    MAX(a.d = DATE_ADD(s.d0, INTERVAL 7 DAY)) AS d7
  FROM starts s LEFT JOIN activity a USING(user_id)
  GROUP BY s.user_id
),
unsub AS (
  SELECT user_id, MAX(is_cancelled) AS unsubscribed              -- ⚠ADJUST: from subscribe_events
  FROM `persona-496908.<dataset>.subscribe_events`               --   e.g. status='cancelled' / cancel_at IS
NOT NULL
  GROUP BY user_id
)
SELECT
  t.engagement_tier,
  COUNT(*)                                          AS users,
  ROUND(AVG(CAST(r.d1 AS INT64)),3)                 AS d1_retention,   -- TARGET METRIC
  ROUND(AVG(CAST(r.d3 AS INT64)),3)                 AS d3_retention,
  ROUND(AVG(CAST(r.d7 AS INT64)),3)                 AS d7_retention,
  ROUND(AVG(CAST(COALESCE(u.unsubscribed,false) AS INT64)),3) AS unsubscribe_rate,
  ROUND(AVG(t.days_completed)/7.0,3)                AS lesson_completion  -- of the 7 days
FROM tiers t
LEFT JOIN ret r USING(user_id)
LEFT JOIN unsub u USING(user_id)
GROUP BY t.engagement_tier
ORDER BY CASE t.engagement_tier WHEN 'high' THEN 1 WHEN 'low' THEN 2 ELSE 3 END;

-- READ IT LIKE THIS: the release "worked" only if there is a smooth dose-response
-- (high < low < none on unsubscribe; high > low > none on D1/D3/D7) that survives the
-- baseline/hold-out check in 03. A high-vs-none gap alone can be pure selection.
```

SQL · 03_segment_breakdown

```
-- Part C · Step 3 — Same metrics, cut by segment (age / profession / plan) + the selection guard.
-- ⚠ADJUST names. Quiz answers (age, status/profession, goal) live on the user/subscribe record or
-- in early app_events params — confirm where in 00.

-- 3a) Metrics by demographic segment × engagement tier
WITH base AS (
  SELECT t.user_id, t.engagement_tier,
         s.age, s.profession, s.plan,                -- ⚠ADJUST source of quiz demographics
         r.d1, r.unsubscribed, t.days_completed
  FROM `persona-496908.<dataset>.v_challenge_metrics_user` t   -- ⚠ a per-user view joining 01+02
  LEFT JOIN `persona-496908.<dataset>.subscribe_events` s USING(user_id)
  LEFT JOIN `persona-496908.<dataset>.v_challenge_ret` r USING(user_id)
)
SELECT age, plan, engagement_tier,
       COUNT(*) users,
       ROUND(AVG(CAST(d1 AS INT64)),3) d1_retention,
       ROUND(AVG(CAST(unsubscribed AS INT64)),3) unsubscribe_rate
FROM base
GROUP BY age, plan, engagement_tier
ORDER BY age, plan, engagement_tier;

-- 3b) THE SELECTION GUARD — dose-response: does each EXTRA challenge day predict lower unsubscribe?
-- If the slope is smooth and monotonic, the effect is more plausibly causal than pure self-selection.
SELECT days_completed,
       COUNT(*) users,
       ROUND(AVG(CAST(unsubscribed AS INT64)),3) unsubscribe_rate,
       ROUND(AVG(CAST(d1 AS INT64)),3) d1_retention
FROM `persona-496908.<dataset>.v_challenge_metrics_user`        -- ⚠ per-user view
GROUP BY days_completed
ORDER BY days_completed;

-- 3c) IDEAL (if a hold-out or pre-launch cohort exists): compare Challenge-exposed vs not,
-- matched on entry motivation (quiz time_goal / goal). If no hold-out exists, the #1
-- recommendation for the next release is to ship the Challenge as a proper A/B, not 100%.
```

SQL · 10_explore_events

```
-- Stage 1 · probe queries — run these next, paste results back.
-- They tell me the full event list (incl. unsubscribe + any challenge-complete/streak events)
-- and exactly how Challenge participation is logged, so I can build the real tier metrics.

-- Q1 — every event type + volume in app_events
SELECT event_name, COUNT(*) AS n
FROM `persona-496908.sql_assessment.app_events`
GROUP BY event_name
ORDER BY n DESC;

-- Q2 — how is the Challenge logged? (path vs course vs category)
SELECT event_name,
       REGEXP_EXTRACT(path, r'/skills/challenges/(\d+)') AS challenge_id,
       course_id, course_name, category_name, mechanic_name,
       COUNT(*) AS n
FROM `persona-496908.sql_assessment.app_events`
WHERE path LIKE '/skills/challenges/%'
   OR LOWER(event_name) LIKE '%challenge%'
   OR LOWER(IFNULL(category_name,'')) LIKE '%challenge%'
GROUP BY 1,2,3,4,5,6
ORDER BY n DESC
LIMIT 200;

-- Q3 — do challenge lessons carry a distinct course/category? peek at raw challenge rows
SELECT event_name, user_id, timestamp, path, course_id, course_name,
       lesson_id, lesson_name, lesson_order, category_name, mechanic_name, status
FROM `persona-496908.sql_assessment.app_events`
WHERE path LIKE '/skills/challenges/%' OR LOWER(event_name) LIKE '%challenge%'
ORDER BY timestamp
LIMIT 50;

-- Q4 — the unsubscribe event (which event_name carries unsubscribe_reason?)
SELECT event_name, unsubscribe_reason, COUNT(*) AS n
FROM `persona-496908.sql_assessment.app_events`
WHERE unsubscribe_reason IS NOT NULL
GROUP BY 1,2
ORDER BY n DESC
LIMIT 100;
```

SQL · 11_challenge_funnel

```
-- Stage 1 · Part C — first real query. Run this, paste the one-row result back.
-- A compact engagement funnel using the confirmed event names (Events Convention).
-- It returns distinct-user counts so we can see the shape before building tier metrics.

SELECT
  COUNT(DISTINCT IF(event_name='pr_webapp_launch_first_time',             user_id, NULL)) AS registered,
  COUNT(DISTINCT IF(event_name='pr_webapp_challenge_view',               user_id, NULL)) AS
challenge_viewed,
  COUNT(DISTINCT IF(event_name='pr_webapp_challenge_start',              user_id, NULL)) AS
challenge_started,
  COUNT(DISTINCT IF(event_name='pr_webapp_challenge_join',               user_id, NULL)) AS
challenge_joined,
  COUNT(DISTINCT IF(event_name='pr_webapp_lesson_completed',            user_id, NULL)) AS
completed_any_lesson,
  COUNT(DISTINCT IF(event_name='pr_webapp_challenge_certificate_download',user_id, NULL)) AS
got_challenge_certificate,
  COUNT(DISTINCT IF(event_name='pr_webapp_challenge_csat_click',          user_id, NULL)) AS
gave_challenge_csat,
  COUNT(DISTINCT IF(event_name='pr_webapp_unsubscribed',                 user_id, NULL)) AS unsubscribed,
  COUNT(DISTINCT user_id)                                                              AS all_users,
  MIN(timestamp) AS first_event, MAX(timestamp) AS last_event
FROM `persona-496908.sql_assessment.app_events`;

-- After this: I'll write query 12 = engagement tiers (high/low/didn't-start) and the
-- D1/D3/D7 + unsubscribe + completion + CSAT comparison, per your definitions.
```

SQL · 12_challenge338_user_table

```
-- 12 · Per-user table for the 7-Day Claude Challenge (id 338).
-- RUN this, then "Save results ▸ CSV" and drop the file in part-c-release-verdict/data/.
-- One row per user who STARTED challenge 338. I'll do the tier / D1-D3-D7 / segment analysis locally.
-- (If it throws an error, paste me the error text — I can't test against your DB.)

WITH ch338 AS (                         -- every event belonging to challenge 338
  SELECT user_id, event_name, timestamp, DATE(timestamp) AS d, csat_score
  FROM `persona-496908.sql_assessment.app_events`
  WHERE course_id = 338 OR REGEXP_CONTAINS(IFNULL(path,''), r'challenges/338')
),
starter AS (                            -- per-user challenge-338 engagement + outcomes
  SELECT
    user_id,
    MIN(IF(event_name='pr_webapp_challenge_start', d, NULL))              AS start_date,
    COUNT(DISTINCT d)                                                     AS active_challenge_days,
    COUNTIF(event_name='pr_webapp_lesson_completed')                     AS lessons_completed,
    MAX(IF(event_name='pr_webapp_course_completed',1,0))                 AS completed_course,
    MAX(IF(event_name IN ('pr_webapp_challenge_certificate_download',
                          'pr_webapp_challenge_certificate_share'),1,0))  AS got_certificate,
    COUNTIF(event_name IN ('pr_webapp_lesson_csat_click','pr_webapp_challenge_csat_click')) AS csat_n,
    AVG(IF(event_name IN ('pr_webapp_lesson_csat_click','pr_webapp_challenge_csat_click'),
           csat_score, NULL))                                             AS avg_csat
  FROM ch338
  GROUP BY user_id
),
act AS (  SELECT user_id, DATE(timestamp) AS d
          FROM `persona-496908.sql_assessment.app_events` GROUP BY 1,2 ),
uns AS (  SELECT user_id, ANY_VALUE(unsubscribe_reason) AS unsub_reason
          FROM `persona-496908.sql_assessment.app_events`
          WHERE event_name='pr_webapp_unsubscribed' GROUP BY 1 ),
seg AS (  SELECT user_id, ANY_VALUE(age) age, ANY_VALUE(gender) gender, ANY_VALUE(goal) goal,
                 ANY_VALUE(status) status, ANY_VALUE(subscription) plan
          FROM `persona-496908.sql_assessment.subscribe_events` GROUP BY 1 )
SELECT
  s.user_id, s.start_date, s.active_challenge_days, s.lessons_completed,
  s.completed_course, s.got_certificate, s.csat_n, ROUND(s.avg_csat,2) AS avg_csat,
  MAX(IF(a.d = DATE_ADD(s.start_date, INTERVAL 1 DAY),1,0)) AS ret_d1,
  MAX(IF(a.d = DATE_ADD(s.start_date, INTERVAL 3 DAY),1,0)) AS ret_d3,
  MAX(IF(a.d = DATE_ADD(s.start_date, INTERVAL 7 DAY),1,0)) AS ret_d7,
  IF(u.user_id IS NULL, 0, 1) AS unsubscribed, u.unsub_reason,
  g.age, g.gender, g.goal, g.status, g.plan
FROM starter s
LEFT JOIN act a ON a.user_id = s.user_id
LEFT JOIN uns u ON u.user_id = s.user_id
LEFT JOIN seg g ON g.user_id = s.user_id
WHERE s.start_date IS NOT NULL          -- only users who actually STARTED the challenge
GROUP BY s.user_id, s.start_date, s.active_challenge_days, s.lessons_completed,
         s.completed_course, s.got_certificate, s.csat_n, s.avg_csat,
         u.user_id, u.unsub_reason, g.age, g.gender, g.goal, g.status, g.plan
ORDER BY s.start_date;
```

SQL · 13_comparison_groups

```
-- 13 · THE COMPARISON QUERY — one row per user, takers AND non-takers of Challenge 338.
-- This is the missing third of Part C Task 1: "high / low / didn't take it".
-- Run it once, then "SAVE RESULTS ▸ CSV (local file)" into part-c-release-verdict/data/.
-- ~10,000 rows (small enough for the local-file download). All tiering + stats done locally in Python.
-- If it errors, paste me the error text verbatim.

WITH
bounds AS (                                   -- last day present in the data (for window correction)
  SELECT MAX(DATE(timestamp)) AS last_data_date
  FROM `persona-496908.sql_assessment.app_events`
),

ev AS (                                       -- every app event + a flag for "belongs to challenge 338"
  SELECT
    user_id,
    event_name,
    DATE(timestamp) AS d,
    csat_score,
    unsubscribe_reason,
    country_code,
    (course_id = 338 OR REGEXP_CONTAINS(IFNULL(path,''), r'challenges/338')) AS ch338
  FROM `persona-496908.sql_assessment.app_events`
),

per_user AS (                                 -- one row per user: engagement, outcomes, challenge-338
detail
  SELECT
    user_id,
    MIN(d)                                                          AS first_seen,
    COUNT(DISTINCT d)                                               AS active_days_total,
    ANY_VALUE(country_code)                                         AS country_code,

    -- overall product engagement (all courses, not just the challenge)
    COUNTIF(event_name = 'pr_webapp_lesson_completed')              AS lessons_completed_all,
    MAX(IF(event_name = 'pr_webapp_onboarding_finished', 1, 0))     AS finished_onboarding,
    MAX(IF(event_name = 'pr_webapp_personal_plan_started', 1, 0))   AS started_personal_plan,

    -- outcome metrics
    MAX(IF(event_name = 'pr_webapp_unsubscribed', 1, 0))            AS unsubscribed,
    ANY_VALUE(IF(event_name = 'pr_webapp_unsubscribed', unsubscribe_reason, NULL)) AS unsub_reason,
    COUNTIF(event_name IN ('pr_webapp_lesson_csat_click',
                           'pr_webapp_module_csat_click',
                           'pr_webapp_challenge_csat_click'))       AS csat_n_all,
    AVG(IF(event_name IN ('pr_webapp_lesson_csat_click',
                          'pr_webapp_module_csat_click',
                          'pr_webapp_challenge_csat_click'), csat_score, NULL)) AS avg_csat_all,

    -- exposure to the challenge surface (any challenge — the popup carries no course_id)
    MAX(IF(event_name = 'pr_webapp_challenge_popup_view', 1, 0))    AS saw_challenge_popup,
    MAX(IF(event_name = 'pr_webapp_challenge_popup_click', 1, 0))   AS clicked_challenge_popup,
    MAX(IF(event_name = 'pr_webapp_challenge_view', 1, 0))          AS viewed_any_challenge,
    MAX(IF(event_name = 'pr_webapp_challenge_start', 1, 0))         AS started_any_challenge,

    -- challenge 338 specifically
    MIN(IF(ch338 AND event_name = 'pr_webapp_challenge_start', d, NULL)) AS ch_start_date,
    MAX(IF(ch338 AND event_name = 'pr_webapp_challenge_view', 1, 0))     AS ch_viewed,
    MAX(IF(ch338 AND event_name = 'pr_webapp_challenge_join', 1, 0))     AS ch_joined,
    COUNTIF(ch338 AND event_name = 'pr_webapp_lesson_completed')         AS ch_lessons_completed,
    COUNTIF(ch338 AND event_name = 'pr_webapp_lesson_started')           AS ch_lessons_started,
    MAX(IF(ch338 AND event_name = 'pr_webapp_course_completed', 1, 0))   AS ch_completed_course,
    MAX(IF(ch338 AND event_name IN ('pr_webapp_challenge_certificate_download',
                                    'pr_webapp_challenge_certificate_share'), 1, 0)) AS ch_certificate,
    COUNTIF(ch338 AND event_name IN ('pr_webapp_lesson_csat_click',
                                     'pr_webapp_challenge_csat_click'))  AS ch_csat_n,
```

```
    AVG(IF(ch338 AND event_name IN ('pr_webapp_lesson_csat_click',
                                    'pr_webapp_challenge_csat_click'), csat_score, NULL)) AS ch_avg_csat
  FROM ev
  GROUP BY user_id
),

anchored AS (                                 -- day 0 for retention: challenge start for takers, first app
day for everyone else
  SELECT
    p.*,
    COALESCE(p.ch_start_date, p.first_seen) AS anchor_date
  FROM per_user p
),

days AS (                                     -- distinct active days per user
  SELECT DISTINCT user_id, DATE(timestamp) AS d
  FROM `persona-496908.sql_assessment.app_events`
),

ret AS (                                      -- retention off BOTH anchors, so takers and non-takers are
comparable
  SELECT
    a.user_id,
    MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 1 DAY), 1, 0)) AS ret_d1,
    MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 3 DAY), 1, 0)) AS ret_d3,
    MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 7 DAY), 1, 0)) AS ret_d7,
    MAX(IF(dd.d = DATE_ADD(a.first_seen,  INTERVAL 1 DAY), 1, 0)) AS fs_ret_d1,
    MAX(IF(dd.d = DATE_ADD(a.first_seen,  INTERVAL 3 DAY), 1, 0)) AS fs_ret_d3,
    MAX(IF(dd.d = DATE_ADD(a.first_seen,  INTERVAL 7 DAY), 1, 0)) AS fs_ret_d7
  FROM anchored a
  JOIN days dd USING (user_id)
  GROUP BY a.user_id
),

seg AS (                                      -- buyer segmentation (NULL = no purchase row for this user)
  SELECT
    user_id,
    ANY_VALUE(age)          AS age,
    ANY_VALUE(gender)       AS gender,
    ANY_VALUE(goal)         AS goal,
    ANY_VALUE(status)       AS status,
    ANY_VALUE(subscription) AS plan,
    MIN(DATE(timestamp))    AS subscribe_date
  FROM `persona-496908.sql_assessment.subscribe_events`
  GROUP BY user_id
)

SELECT
  a.user_id,

  -- the comparison group (I tier "high vs low" locally from ch_lessons_completed)
  CASE
    WHEN a.ch_start_date IS NOT NULL THEN 'took_338'
    WHEN a.ch_joined = 1             THEN 'joined_338_never_started'
    WHEN a.ch_viewed  = 1            THEN 'viewed_338_never_joined'
    WHEN a.started_any_challenge = 1 THEN 'took_other_challenge'
    ELSE                                  'no_challenge'
  END AS group_338,

  a.first_seen,
  a.anchor_date,
  DATE_DIFF(b.last_data_date, a.anchor_date, DAY) AS days_observed,   -- <7 ⇒ D7 not observable

  a.ch_start_date, a.ch_lessons_started, a.ch_lessons_completed,
  a.ch_completed_course, a.ch_certificate, a.ch_csat_n, ROUND(a.ch_avg_csat, 2) AS ch_avg_csat,

  a.lessons_completed_all, a.active_days_total,
  a.finished_onboarding, a.started_personal_plan,
  a.saw_challenge_popup, a.clicked_challenge_popup,
```

```
  a.viewed_any_challenge, a.started_any_challenge,

  r.ret_d1, r.ret_d3, r.ret_d7,
  r.fs_ret_d1, r.fs_ret_d3, r.fs_ret_d7,

  a.unsubscribed, a.unsub_reason,
  a.csat_n_all, ROUND(a.avg_csat_all, 2) AS avg_csat_all,

  s.age, s.gender, s.goal, s.status, s.plan, s.subscribe_date,
  IF(s.user_id IS NULL, 0, 1) AS has_subscription,
  a.country_code
FROM anchored a
CROSS JOIN bounds b
LEFT JOIN ret r ON r.user_id = a.user_id
LEFT JOIN seg s ON s.user_id = a.user_id
ORDER BY a.user_id;
```

Code · Verification — re-derives every published figure

```
"""QA: re-derive every load-bearing number in the write-ups straight from the CSVs and assert it.

If a number in the markdown ever drifts from the data, this fails loudly.
Run:  python3 05_qa.py
"""
import csv
import os
from collections import Counter
from lib import load, wilson, two_prop_z, DATA

rows = load()
N = len(rows)
takers = [r for r in rows if r["took"]]
nont = [r for r in rows if not r["took"]]
fails, checks = [], 0


def check(label, actual, expected, tol=0.0005):
    # NB tolerances are deliberately tight: a 0.001 tolerance once hid a 59.557 -> '59.5%' mis-rounding.
    """expected may be an int/float; tol is absolute."""
    global checks
    checks += 1
    ok = abs(actual - expected) <= tol
    if not ok:
        fails.append(f"{label}: computed {actual!r}, document says {expected!r}")
    print(f"  {'ok ' if ok else 'FAIL'} {label:<58} {actual:.4f}  (doc: {expected})")


def rate(g, k):
    return sum(r[k] for r in g) / len(g)


print("== cohort shape ==")
check("subscribers in cohort", N, 9956, 0)
check("challenge-338 starters", len(takers), 1111, 0)
check("non-takers", len(nont), 8845, 0)
check("all rows have a subscription", sum(r["has_subscription"] for r in rows), 9956, 0)

print("\n== exposure ladder (Part C §1) ==")
check("saw popup", sum(r["saw_challenge_popup"] for r in rows), 3140, 0)
check("saw popup %", sum(r["saw_challenge_popup"] for r in rows) / N, 0.315, 0.0005)
check("clicked popup", sum(r["clicked_challenge_popup"] for r in rows), 2970, 0)
check("viewed a challenge", sum(r["viewed_any_challenge"] for r in rows), 2109, 0)
check("started 338 %", len(takers) / N, 0.112, 0.0005)
check("never reached a surface",
      sum(1 for r in rows if not r["saw_challenge_popup"] and not r["viewed_any_challenge"]), 6716, 0)
check("never reached a surface %",
      sum(1 for r in rows if not r["saw_challenge_popup"] and not r["viewed_any_challenge"]) / N, 0.675,
0.0005)

print("\n== inside the challenge (Part C §2) ==")
for k, exp in [(1, 612), (2, 308), (3, 158), (8, 47)]:
    check(f"completed >= {k} lessons", sum(1 for r in takers if r["ch_lessons_completed"] >= k), exp, 0)
check("zero lessons %", sum(1 for r in takers if r["ch_lessons_completed"] == 0) / len(takers), 0.449,
0.0005)
check("lesson1->2 survival", 308 / 612, 0.503, 0.001)
check("completed the course", sum(r["ch_completed_course"] for r in takers), 48, 0)
check("certificate", sum(r["ch_certificate"] for r in takers), 23, 0)

print("\n== tiers + headline table (Part C §4) ==")
high = [r for r in takers if r["ch_lessons_completed"] >= 3]
low = [r for r in takers if r["ch_lessons_completed"] < 3]
```

```
check("HIGH n", len(high), 158, 0)
check("LOW n", len(low), 953, 0)
check("HIGH D1", rate(high, "fs_ret_d1"), 0.696, 0.001)
check("LOW D1", rate(low, "fs_ret_d1"), 0.584, 0.001)
check("DIDNT-TAKE D1", rate(nont, "fs_ret_d1"), 0.217, 0.001)
check("HIGH D3", rate(high, "fs_ret_d3"), 0.570, 0.001)
check("LOW D3", rate(low, "fs_ret_d3"), 0.344, 0.001)
check("DIDNT-TAKE D3", rate(nont, "fs_ret_d3"), 0.100, 0.001)
for label, g, exp, expn in [("HIGH", high, 0.391, 133), ("LOW", low, 0.180, 737), ("DIDNT-TAKE", nont,
0.069, 6345)]:
    obs = [r for r in g if r["fs_days_observed"] >= 7]
    check(f"{label} D7", rate(obs, "fs_ret_d7"), exp, 0.001)
    check(f"{label} D7 denominator", len(obs), expn, 0)
check("HIGH unsub", rate(high, "unsubscribed"), 0.114, 0.001)
check("LOW unsub", rate(low, "unsubscribed"), 0.114, 0.001)
check("DIDNT-TAKE unsub", rate(nont, "unsubscribed"), 0.143, 0.001)

print("\n== the decisive test (Part C §5 Test A) ==")
ctrl = [r for r in rows if r["group_338"] in ("joined_338_never_started", "viewed_338_never_joined")]
check("takers D1", rate(takers, "fs_ret_d1"), 0.600, 0.001)
check("control n", len(ctrl), 994, 0)
check("control D1 (the 59.6%)", rate(ctrl, "fs_ret_d1"), 0.596, 0.0005)
d, z, p = two_prop_z(sum(r["fs_ret_d1"] for r in takers), len(takers),
                     sum(r["fs_ret_d1"] for r in ctrl), len(ctrl))
check("D1 effect (pts)", d, 0.005, 0.001)
check("D1 effect p-value", p, 0.823, 0.005)
d2, _, p2 = two_prop_z(sum(r["unsubscribed"] for r in takers), len(takers),
                       sum(r["unsubscribed"] for r in ctrl), len(ctrl))
check("unsub effect (pts)", d2, -0.023, 0.001)
check("unsub effect p-value", p2, 0.119, 0.005)
check("joined-never-started D1", rate([r for r in rows if r["group_338"] == "joined_338_never_started"],
"fs_ret_d1"), 0.604, 0.001)
check("viewed-never-joined D1", rate([r for r in rows if r["group_338"] == "viewed_338_never_joined"],
"fs_ret_d1"), 0.593, 0.001)

print("\n== immortal time + dose-response (Test B / C) ==")
lag = Counter(r["days_to_challenge"] for r in takers)
check("start on day 0 %", lag[0] / len(takers), 0.239, 0.001)
oneday = [r for r in takers if r["active_challenge_days"] == 1]
hi = [r for r in oneday if r["ch_lessons_completed"] >= 2]
lo = [r for r in oneday if r["ch_lessons_completed"] <= 1]
d3, _, p3 = two_prop_z(sum(r["fs_ret_d1"] for r in hi), len(hi),
                       sum(r["fs_ret_d1"] for r in lo), len(lo))
check("day-0 dose effect (pts, negative)", d3, -0.092, 0.001)
check("day-0 dose p-value", p3, 0.048, 0.002)

print("\n== the mechanic (Test D) ==")
fin = [r for r in takers if r["ch_completed_course"] == 1]
check("finished in 1 day", sum(1 for r in fin if r["active_challenge_days"] == 1), 13, 0)
check("finished in 1 day %", sum(1 for r in fin if r["active_challenge_days"] == 1) / len(fin), 0.271,
0.001)
check("finished in <=2 days %", sum(1 for r in fin if r["active_challenge_days"] <= 2) / len(fin), 0.458,
0.001)
did = [r for r in takers if r["ch_lessons_completed"] >= 1]
check("returned for a 2nd challenge day %",
      sum(1 for r in did if r["active_challenge_days"] >= 2) / len(did), 0.368, 0.001)
check("mean lessons per active challenge day",
      sum(r["ch_lessons_completed"] / max(1, r["active_challenge_days"]) for r in did) / len(did), 1.46,
0.005)

print("\n== the metric that would have fooled the team (§6) ==")
check("takers D1, challenge-anchored", rate(takers, "ret_d1"), 0.398, 0.001)
check("cohort D1, first-app-day", rate([r for r in rows if r["fs_days_observed"] >= 1], "fs_ret_d1"), 0.260,
0.001)

print("\n== CSAT inversion (§8) ==")


def csat(g):
```

```
    v = [r["avg_csat_all"] for r in g if r["csat_n_all"] > 0]
    return sum(v) / len(v), len(v)


for label, g, exp in [("never took", nont, 3.74), ("LOW", low, 3.53), ("HIGH", high, 3.33), ("finishers",
fin, 3.10)]:
    m, n = csat(g)
    check(f"CSAT {label}", m, exp, 0.005)
chr_ = [r for r in takers if r["ch_csat_n"] > 0]
check("CSAT of challenge-338 content", sum(r["ch_avg_csat"] for r in chr_) / len(chr_), 3.25, 0.005)
check("same users, all content", sum(r["avg_csat_all"] for r in chr_ if r["csat_n_all"] > 0) /
      len([r for r in chr_ if r["csat_n_all"] > 0]), 3.45, 0.005)

print("\n== product-wide baselines (the thesis number) ==")
check("never completed a lesson anywhere %",
      sum(1 for r in rows if r["lessons_completed_all"] == 0) / N, 0.502, 0.001)
check("cohort unsubscribe %", rate(rows, "unsubscribed"), 0.140, 0.001)

print("\n== demographics (Part A + the PM's claim) ==")
check("male %", sum(1 for r in rows if r["gender"] == "Male") / N, 0.597, 0.001)
check("female %", sum(1 for r in rows if r["gender"] == "Female") / N, 0.384, 0.001)
check("aged 45+ (clean buckets) %", sum(1 for r in rows if r["age"] in ("45-54", "55+")) / N, 0.578, 0.001)
check("aged 45+ incl. legacy bucket %",
      sum(1 for r in rows if r["age"] in ("45-54", "55+", "45+")) / N, 0.601, 0.001)
check("men 55+ count", sum(1 for r in rows if r["gender"] == "Male" and r["age"] == "55+"), 2120, 0)
check("men 55+ %", sum(1 for r in rows if r["gender"] == "Male" and r["age"] == "55+") / N, 0.213, 0.001)
check("men 35-54 %", sum(1 for r in rows if r["gender"] == "Male" and r["age"] in ("35-44", "45-54")) / N,
0.249, 0.001)
for a, exp in [("18-24", 0.311), ("25-34", 0.197), ("35-44", 0.145), ("45-54", 0.095), ("55+", 0.097)]:
    check(f"unsub, age {a}", rate([r for r in rows if r["age"] == a], "unsubscribed"), exp, 0.001)
for gl, exp in [("Work faster", 0.109), ("Feel more confident with AI", 0.107), ("Get a quick side hustle",
0.372)]:
    check(f"unsub, goal '{gl}'", rate([r for r in rows if r["goal"] == gl], "unsubscribed"), exp, 0.001)

print("\n== plans (Part D validation) ==")
pc = Counter(r["plan"] for r in rows)
check("1Week share", pc["1Week"] / N, 0.102, 0.001)
check("4Week share incl. special", (pc["4Week"] + pc["4Week_special"]) / N, 0.646, 0.001)
check("12Week share", pc["12Week"] / N, 0.251, 0.001)
check("1Week unsub", rate([r for r in rows if r["plan"] == "1Week"], "unsubscribed"), 0.346, 0.001)
check("4Week unsub", rate([r for r in rows if r["plan"] == "4Week"], "unsubscribed"), 0.122, 0.001)
check("12Week unsub", rate([r for r in rows if r["plan"] == "12Week"], "unsubscribed"), 0.106, 0.001)
blended = (0.102 * 60.42 + 0.646 * 123.70 + 0.251 * 162.15) / 0.999
check("observed-mix blended net LTV", blended, 126.90, 0.02)

print("\n== unsubscribe reasons (§8) ==")
uns = [r for r in rows if r["unsubscribed"]]
check("total unsubscribed", len(uns), 1393, 0)
invol = {"hard_decline", "Mastercard Alert", "dispute", "paypal_refund", "Visa CDRN",
         "fraud", "Merchanto visa", "payment_refunded", "fraud_reported"}
ni = sum(1 for r in uns if r["unsub_reason"] in invol)
check("payment/chargeback/dispute unsubs", ni, 157, 0)
check("… as % of unsubs", ni / len(uns), 0.113, 0.001)

print("\n== event catalogue (the 'not observable here' claim) ==")
# The verdict rests on primary metrics being uncomputable from this dataset. That is a claim
# about the event catalogue, not about the user table, so it needs its own check or it would
# be the one published number nothing re-derives.
CAT = os.path.join(DATA, "script_job_ee3720964a3e4c3ac51fe64306b65890_0.csv")
cat = {r["event_name"]: int(r["n"]) for r in csv.DictReader(open(CAT))}
check("event catalogue rows (file integrity)", len(cat), 67, 0)
check("subscription_renewed events", cat.get("pr_webapp_subscription_renewed", 0), 17, 0)
check("… per subscriber", cat.get("pr_webapp_subscription_renewed", 0) / N, 0.0017, 0.0001)

print("\n" + "=" * 70)
if fails:
    print(f"{len(fails)} of {checks} CHECKS FAILED:")
    for f in fails:
        print("   -", f)
```

```
    raise SystemExit(1)
print(f"All {checks} checks passed — every documented number matches the data.")
```

Code · Part D — the LTV and A/B model

```
#!/usr/bin/env python3
"""
Part D — Subscription economics model (Jobescape PM take-home).

Task 1: Total Net LTV over a one-year horizon (per plan gross+net, blended).
Task 2: A/B plan-upgrade break-even (4-week -> 12-week upgrade replacing 2nd upsell).

Design decisions / assumptions (documented explicitly, per brief):
- Cohort of 1 subscriber per plan; everyone pays the intro (survival S0 = 1).
- Cmn = P(pay period n | survived to period m). Survival S_k = product of C01..C_{k-1,k}.
- Period 0 = intro (pays Intro Price). Periods 1..12 = recurring (pay Recurring Price),
  weighted by survival. The data supplies exactly 12 renewal transitions (C01..C1112).
- Payment-provider fee = 12% on ALL collected revenue (intro, recurring, upsells). Net = Gross * (1-0.12).
- Upsells are one-time at checkout; expected value = conv_rate * price, applied once to every subscriber.
- HORIZON: two readings are computed because the brief says "one-year horizon" but gives 12
  transitions per plan:
    (A) FULL-CURVE  : apply all 12 transitions to every plan (matches the data given; for the
                      4-week plan 13 periods x 4wk = exactly 52 weeks). PRIMARY.
    (B) STRICT-52WK : count only payments landing within 52 calendar weeks
                      (12-week plan -> only 4 recurring periods; others unchanged). SENSITIVITY.
"""

FEE = 0.12

plans = {
    "1-WEEK":  dict(mix=0.10, intro=6.93,  rec=39.99, rec_wk=4,  intro_wk=1,
                    C=[.55,.50,.60,.75,.80,.80,.80,.80,.80,.80,.80,.80],
                    up1=(0.30,1.99),  up2=(0.12,0.99)),
    "4-WEEK":  dict(mix=0.70, intro=19.99, rec=39.99, rec_wk=4,  intro_wk=4,
                    C=[.67,.65,.70,.75,.80,.80,.80,.80,.80,.80,.80,.80],
                    up1=(0.30,69.99), up2=(0.12,29.99)),
    "12-WEEK": dict(mix=0.20, intro=39.99, rec=62.99, rec_wk=12, intro_wk=12,
                    C=[.64,.57,.65,.75,.75,.75,.75,.75,.75,.75,.75,.75],
                    up1=(0.30,69.99), up2=(0.12,29.99)),
}

def survivals(C):
    s, acc = [], 1.0
    for c in C:
        acc *= c
        s.append(acc)          # s[k-1] = survival to recurring period k
    return s

def periods_within_year(intro_wk, rec_wk):
    """Recurring payments CHARGED within the 52-week horizon (strict reading).
    Recurring period k is charged at week intro_wk + (k-1)*rec_wk; count it if that
    charge falls inside the year (LTV = cash collected, so charge date is what counts,
    not whether the service period extends past week 52). Capped at the 12 given transitions.
    -> 12-week plan: charges at wk 12/24/36/48 -> 4 periods; 4-week & 1-week -> 12."""
    n = 0
    for k in range(1, 13):
        if intro_wk + (k - 1) * rec_wk < 52:
            n += 1
        else:
            break
    return n

def recurring_revenue(p, mode):
    s = survivals(p["C"])
    if mode == "full":
        k = 12
    else:
        k = periods_within_year(p["intro_wk"], p["rec_wk"])
    return p["rec"] * sum(s[:k]), k
```

```
def upsell_ev(p):
    return p["up1"][0]*p["up1"][1] + p["up2"][0]*p["up2"][1]

def plan_ltv(p, mode):
    rec_rev, k = recurring_revenue(p, mode)
    sub_gross = p["intro"] + rec_rev
    up = upsell_ev(p)
    gross = sub_gross + up
    return dict(k=k, intro=p["intro"], rec_rev=rec_rev, upsell=up,
                gross=gross, net=gross*(1-FEE))

print("="*72)
print("TASK 1 — LTV per plan (1-year horizon).  Fee = 12%.")
for mode, label in [("full","(A) FULL-CURVE  [PRIMARY]"), ("strict","(B) STRICT-52WK [sensitivity]")]:
    print(f"\n--- Horizon reading {label} ---")
    print(f"{'plan':8} {'mix':>5} {'recPds':>6} {'intro':>7} {'recRev':>8} {'upsell':>7} {'GROSS':>8}
{'NET':>8}")
    blended = 0.0
    for name, p in plans.items():
        r = plan_ltv(p, mode)
        blended += p["mix"]*r["net"]
        print(f"{name:8} {p['mix']*100:4.0f}% {r['k']:6d} {r['intro']:7.2f} {r['rec_rev']:8.2f}
{r['upsell']:7.2f} {r['gross']:8.2f} {r['net']:8.2f}")
    print(f"{'BLENDED NET LTV (mix-weighted)':>55}: ${blended:6.2f}")

print("\n"+"="*72)
print("TASK 2 — A/B: 4-week buyer offered Plan Upgrade ($49.99) INSTEAD of 2nd upsell.")
print("Upgrade buyer becomes a 12-week subscriber (12-week recurring economics).")
UP_PRICE = 49.99
for mode, label in [("full","FULL-CURVE"), ("strict","STRICT-52WK")]:
    p4, p12 = plans["4-WEEK"], plans["12-WEEK"]
    recRev4,_  = recurring_revenue(p4, mode)
    recRev12,_ = recurring_revenue(p12, mode)
    up2_4 = p4["up2"][0]*p4["up2"][1]                # expected 2nd-upsell value being replaced
    # Break-even p*: p*(UP_PRICE + recRev12 - recRev4) = up2_4   (intro & 1st upsell cancel; fee uniform)
    denom = UP_PRICE + recRev12 - recRev4
    p_star = up2_4 / denom
    print(f"\n--- {label} ---")
    print(f"  recurring rev 4-week  = ${recRev4:6.2f}")
    print(f"  recurring rev 12-week = ${recRev12:6.2f}  (delta vs 4wk = ${recRev12-recRev4:+.2f})")
    print(f"  2nd-upsell value replaced (0.12*29.99) = ${up2_4:.2f}")
    print(f"  each upgrade adds ${denom:.2f} gross; cannibalises ${up2_4:.2f}")
    print(f"  >> BREAK-EVEN upgrade conversion p* = {p_star*100:.2f}%")
    # sample scenarios
    ctrl_gross = p4["intro"] + recRev4 + p4["up1"][0]*p4["up1"][1] + up2_4
    print(f"  control 4-week gross LTV = ${ctrl_gross:.2f} (net ${ctrl_gross*(1-FEE):.2f})")
    for p in (0.05,0.10,0.20,0.30):
        test_gross = (p4["intro"] + p4["up1"][0]*p4["up1"][1]
                      + (1-p)*recRev4 + p*(UP_PRICE + recRev12))
        lift = (test_gross-ctrl_gross)/ctrl_gross*100
        print(f"     p={p*100:4.0f}%  test gross LTV ${test_gross:7.2f}  net ${test_gross*(1-FEE):7.2f}
lift {lift:+5.1f}%")
```

Code · Part A — the segment clustering

```
#!/usr/bin/env python3
"""Part A — derive audience segments from the quiz data instead of asserting them.

WHY THIS EXISTS
    `segment_sizer.py` applied a hand-written if/elif rule chain: the six segments were authored
    from a jobs-to-be-done framework and then counted, with unmatched users falling through to a
    default. That is a taxonomy, not a segmentation — the structure came from the analyst, not the
    data, so it cannot be falsified. This script asks the opposite question: if nobody told the
    data what the segments were, how many groups are actually there, and what are they?

METHOD
    k-modes (Huang, 1998) on the categorical quiz answers.
      · Distance is Hamming — the count of questions two users answered differently.
      · Centroids are per-attribute MODES, not means. k-means is wrong here: a mean over nominal
        levels ("35-44", "Business owner") has no meaning, and one-hot + Euclidean silently
        asserts that every pair of levels is equidistant, which is the assumption under test.
      · Missing answers are kept as their own level. In a 26-step quiz funnel, *not answering*
        is behaviour (where someone dropped), not absent data — discarding it would bias the
        sample toward finishers.
      · k chosen by the elbow in within-cluster cost, cross-checked against interpretability.
      · Multiple restarts per k; the best-cost run wins. Seeded, so it reproduces exactly.

READ THE OUTPUT AS
    cluster profiles are reported by LIFT (share inside the cluster ÷ share in the population),
    not by raw share. The modal answer of a cluster is usually just the modal answer of the whole
    population; what makes a cluster a cluster is what it over-represents.
"""
import collections
import csv
import os
import random
import sys

random.seed(42)
csv.field_size_limit(10 ** 7)

HERE = os.path.dirname(os.path.abspath(__file__))
SRC = os.path.join(HERE, "..", "materials", "quiz-funnel-answers", "Quiz Funnel Answers.csv")

# The questions that carry motive or identity. Chosen for coverage (>25k answers each) and
# because each maps to something a PM could act on. IDs are the `ID` column, not `question_id`
# (which is 0 for every row in this export).
CORE = {
    "391":  "age",
    "357":  "gender",
    "394":  "work_status",
    "331":  "learn_claude_for",
    "390":  "benefit",
    "332":  "ai_experience",
    "336":  "career_fear",
    "335":  "feels_ready",
    "1241": "used_claude",
    "1426": "field",
}
MIN_ANSWERED = 6          # users answering fewer than this are quiz drop-offs, not segments
NA = "(no answer)"


def load():
    """Pivot the long export (one row per user-question) into one row per user."""
    wide = collections.defaultdict(dict)
    with open(SRC, encoding="utf-8-sig") as f:
        for r in csv.DictReader(f):
            qid = r["ID"]
```

```
            if qid in CORE:
                a = r["question_answer"].strip()
                if a:
                    wide[r["epv_user_id"]][CORE[qid]] = a
    cols = list(CORE.values())
    rows = [[u.get(c, NA) for c in cols] for u in wide.values()]
    kept = [r for r in rows if sum(v != NA for v in r) >= MIN_ANSWERED]
    return cols, rows, kept


def cost(rows, modes, assign):
    return sum(sum(a != b for a, b in zip(rows[i], modes[assign[i]])) for i in range(len(rows)))


def kmodes(rows, k, iters=12, restarts=4):
    best = None
    m = len(rows[0])
    for _ in range(restarts):
        modes = [list(r) for r in random.sample(rows, k)]
        assign = [0] * len(rows)
        for _ in range(iters):
            moved = False
            for i, r in enumerate(rows):
                d = [sum(a != b for a, b in zip(r, mo)) for mo in modes]
                new = d.index(min(d))
                if new != assign[i]:
                    assign[i] = new
                    moved = True
            for c in range(k):
                members = [rows[i] for i in range(len(rows)) if assign[i] == c]
                if not members:
                    modes[c] = list(random.choice(rows))
                    continue
                modes[c] = [collections.Counter(mem[j] for mem in members).most_common(1)[0][0]
                            for j in range(m)]
            if not moved:
                break
        c = cost(rows, modes, assign)
        if best is None or c < best[0]:
            best = (c, modes, assign[:])
    return best


def main():
    cols, allrows, rows = load()
    n = len(rows)
    print(f"quiz respondents in export        {len(allrows):,}")
    print(f"answered >= {MIN_ANSWERED} of {len(cols)} core questions  {n:,}  "
          f"({n/len(allrows)*100:.1f}%)  <- the clustering sample\n")

    base = [collections.Counter(r[j] for r in rows) for j in range(len(cols))]

    print("== choosing k: within-cluster Hamming cost ==")
    print("   k     cost   cost/user   improvement")
    prev = None
    results = {}
    for k in range(2, 9):
        c, modes, assign = kmodes(rows, k)
        results[k] = (c, modes, assign)
        imp = f"{(prev-c)/prev*100:5.1f}%" if prev else "    —"
        print(f"  {k:>2}  {c:>7,}   {c/n:>8.3f}   {imp}")
        prev = c

    k = int(sys.argv[1]) if len(sys.argv) > 1 else 6
    c, modes, assign = results[k]
    sizes = collections.Counter(assign)
    print(f"\n== k = {k} · cluster profiles (lift = share in cluster / share in population) ==")
    for cl, sz in sizes.most_common():
        members = [rows[i] for i in range(n) if assign[i] == cl]
```

```
        print(f"\n  ── cluster {cl}  n = {sz:,}  ({sz/n*100:.1f}%)")
        marks = []
        for j, col in enumerate(cols):
            cnt = collections.Counter(m[j] for m in members)
            for val, v in cnt.most_common(3):
                if val == NA:
                    continue
                share = v / sz
                pop = base[j][val] / n
                if pop > 0 and share > .18:
                    marks.append((share / pop, col, val, share))
        for lift, col, val, share in sorted(marks, reverse=True)[:6]:
            print(f"      {lift:4.2f}x  {share*100:4.1f}%  {col:16} {val[:52]}")


if __name__ == "__main__":
    main()
```